

indiv_region_increases

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This script test whether non-EUCA per-capita arrest rates rose more than EUCA per-capita rates in all states and AORs and with all denominators

```
aggregate_data <- function(data,
                             cols_to_keep,
                             pop_col,
                             agg_noneuca = TRUE){

  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(aor, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("aor", "region"))

  # aggregate regions if requested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(pick(-c(n_arrests, pop))) %>%
      summarize(n_arrests = sum(n_arrests),
                pop = sum(pop))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(aor, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # population map
  pop_map <- data_pop %>%
    group_by(aor, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")
}
```

```

# checksum value
checksum <- data_pop %>%
  group_by(aor, region, pop) %>%
  slice_head(n=1) %>%
  ungroup() %>%
  group_by(region) %>%
  summarize(pop_checksum = sum(pop), .groups = "drop")

# aggregate arrests, join population back in
data_pop <- data_pop %>%
  group_by_at(cols_to_keep) %>%
  summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
  left_join(pop_map, by = geo_keys) %>%
  mutate(ratio = n_arrests/pop*1000,
         log_pop = log(pop))

# checksum
total_aggregated <- data_pop %>%
  group_by_at(geo_keys) %>%
  summarize(pop = first(pop), .groups = "drop") %>%
  pull(pop) %>%
  sum(na.rm = TRUE)
total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
if (!(dplyr::near(total_checksum, total_aggregated))) {
  warning("Checksum Failed!")
}

return(data_pop)
}

```

MPI, AOR

```

trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv") %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)

## Rows: 113400 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)

## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

```

```
## i Summaries were computed grouped by aor, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by aor, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aor, region, window, convicted, threat_level, method,
##   alt_method))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, aor)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

##	aor	diff_noneuca	diff_euca	diff
## 1	San Francisco	0.2346120	0.17335922	0.06125274
## 2	Los Angeles	0.9107710	0.78647768	0.12429332
## 3	Boston	0.1222781	-0.04000000	0.16227806
## 4	Seattle	0.1936937	-0.14705882	0.34075252
## 5	Buffalo	0.3210973	-0.03733806	0.35843538
## 6	Baltimore	1.4776536	1.00000000	0.47765363
## 7	New York City	0.9140687	0.22869211	0.68537658
## 8	Newark	2.0290828	1.06896552	0.96011726
## 9	Dallas	8.7422661	6.02696258	2.71530353
## 10	San Diego	6.3094339	2.90393661	3.40549731
## 11	Denver	5.0460829	1.36842105	3.67766190
## 12	Washington	6.0054496	1.50000000	4.50544959
## 13	San Antonio	6.1474507	1.33713376	4.81031689
## 14	Chicago	6.0683572	0.55670103	5.51165619
## 15	Phoenix	6.0034483	0.26666667	5.73678161
## 16	Salt Lake City	8.0924370	2.04000000	6.05243697
## 17	Atlanta	7.1219029	1.04918033	6.07272255
## 18	Philadelphia	7.3134921	0.56756757	6.74592450
## 19	St. Paul	7.3467337	0.42105263	6.92568104
## 20	Houston	9.3981837	2.23455240	7.16363127
## 21	Miami	13.7493495	3.59154930	10.15780023
## 22	Harlingen	15.1463813	2.41050312	12.73587821
## 23	Detroit	14.7916667	0.80769231	13.98397436
## 24	New Orleans	19.5393519	1.66666667	17.87268519
## 25	El Paso	26.7253745	2.78101102	23.94436344

ACS, AOR

```
trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv") %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)
```

```
## Rows: 113400 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by aor, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by aor, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aor, region, window, convicted, threat_level, method,
##   alt_method))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, aor)
```

```
trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

##		aor	diff_noneuca	diff_euca	diff
## 1	San Francisco	0.07236487	0.022717697	0.04964718	
## 2	Boston	0.04465381	-0.006160518	0.05081433	
## 3	Seattle	0.06600311	-0.029794182	0.09579729	
## 4	New York City	0.13661890	0.021340618	0.11527828	
## 5	Los Angeles	0.24110148	0.118801844	0.12229964	
## 6	Buffalo	0.28047689	-0.009438593	0.28991548	

## 7	Newark	0.48553382	0.098818009	0.38671581
## 8	Baltimore	0.58147907	0.157853197	0.42362587
## 9	St. Paul	1.85695596	0.075723156	1.78123281
## 10	San Diego	2.17869500	0.378582335	1.80011266
## 11	Washington	2.05910683	0.258242231	1.80086460
## 12	Philadelphia	2.07534061	0.100763403	1.97457720
## 13	Chicago	2.09238822	0.091771340	2.00061688
## 14	Detroit	2.08525869	0.075251555	2.01000713
## 15	Denver	2.30217098	0.256620311	2.04555067
## 16	Phoenix	2.16946770	0.030055979	2.13941172
## 17	Dallas	3.87335292	1.064152057	2.80920086
## 18	Salt Lake City	3.30795214	0.361899761	2.94605238
## 19	Houston	3.27111880	0.288275067	2.98284374
## 20	San Antonio	3.26119943	0.171425009	3.08977442
## 21	Atlanta	3.42829609	0.217798196	3.21049789
## 22	Miami	3.83072696	0.458699919	3.37202704
## 23	Harlingen	4.77777877	1.225677953	3.55210081
## 24	El Paso	8.95911611	0.421213862	8.53790225
## 25	New Orleans	9.13796146	0.392513973	8.74544748

```

aggregate_data <- function(data,
                             cols_to_keep,
                             pop_col,
                             agg_noneuca = TRUE){

  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(state, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # aggregate regions if requested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA"))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(state, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # checksum value
  checksum <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

```

```

# find geographic keys
geo_keys <- intersect(cols_to_keep, c("state", "region"))

# population map
pop_map <- data_pop %>%
  group_by(state, region, pop) %>%
  slice_head(n = 1) %>%
  ungroup() %>%
  group_by_at(geo_keys) %>%
  summarize(pop = sum(pop), .groups = "drop")

# aggregate arrests, join population back in
data_pop <- data_pop %>%
  group_by_at(cols_to_keep) %>%
  summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
  left_join(pop_map, by = geo_keys) %>%
  mutate(ratio = n_arrests/pop*1000,
         log_pop = log(pop))

# checksum
total_aggregated <- data_pop %>%
  group_by_at(geo_keys) %>%
  summarize(pop = first(pop), .groups = "drop") %>%
  pull(pop) %>%
  sum(na.rm = TRUE)
total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
if (!(dplyr::near(total_checksum, total_aggregated))) {
  warning("Checksum Failed!")
}

return(data_pop)
}

```

MPI, state

```

trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv") %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)

## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)

```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
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## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, state)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, state) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

##	state	diff_noneuca	diff_euca	diff
## 1	MASSACHUSETTS	-0.2485549	-0.23809524	-0.01045968
## 2	RHODE ISLAND	1.0000000	1.00000000	0.00000000
## 3	CONNECTICUT	0.1306818	0.09090909	0.03977273
## 4	OKLAHOMA	0.1188119	0.00000000	0.11881188
## 5	OREGON	0.2720588	0.11764706	0.15441176
## 6	WASHINGTON	0.1144781	-0.29787234	0.41235045
## 7	CALIFORNIA	1.0938059	0.65811966	0.43568629
## 8	NEW YORK	0.7402945	0.14772727	0.59256724
## 9	MARYLAND	1.6061453	1.00000000	0.60614525
## 10	ILLINOIS	1.0575816	0.24242424	0.81515733
## 11	NEW JERSEY	2.0111857	1.06896552	0.94222017
## 12	ALASKA	1.5555556	0.50000000	1.05555556
## 13	HAWAII	1.9642857	0.64285714	1.32142857
## 14	COLORADO	2.1698113	0.55555556	1.61425577
## 15	VERMONT	2.0000000	0.00000000	2.00000000
## 16	DELAWARE	1.1071429	-1.00000000	2.10714286
## 17	PENNSYLVANIA	2.4953704	0.11764706	2.37772331
## 18	MINNESOTA	1.9560440	-0.60000000	2.55604396
## 19	NORTH CAROLINA	4.5046948	1.50000000	3.00469484
## 20	WISCONSIN	4.7000000	1.33333333	3.36666667
## 21	VIRGINIA	5.6088235	1.60000000	4.00882353
## 22	NEVADA	6.8709677	2.60000000	4.27096774
## 23	NEBRASKA	6.4313725	1.50000000	4.93137255
## 24	TEXAS	10.0156087	4.33333333	5.68227541
## 25	ARIZONA	6.0310345	0.26666667	5.76436782
## 26	GEORGIA	6.5438596	0.66666667	5.87719298
## 27	IDAHO	7.6969697	1.25000000	6.44696970
## 28	NEW HAMPSHIRE	7.8750000	0.40000000	7.47500000

```
## 29          UTAH      10.1968504  1.80000000    8.39685039
## 30 DISTRICT OF COLUMBIA    9.5384615  0.75000000    8.78846154
## 31          KANSAS      11.5063291  2.00000000    9.50632911
## 32          FLORIDA      13.7701648  3.60563380   10.16453098
## 33          LOUISIANA     11.9569892  1.00000000   10.95698925
## 34          MAINE       10.0000000 -1.00000000   11.00000000
## 35          INDIANA      14.7983193  3.12500000   11.67331933
## 36          MICHIGAN     12.0634921  0.31250000   11.75099206
## 37          KENTUCKY     12.8035714  0.60000000   12.20357143
## 38          IOWA       13.3409091  0.80000000   12.54090909
## 39          OHIO       16.9135802  1.60000000   15.31358025
## 40 SOUTH CAROLINA     17.8425197  0.92307692   16.91944276
## 41          ARKANSAS     19.0000000  1.12500000   17.87500000
## 42          TENNESSEE    19.2603550  1.27272727   17.98762776
## 43          MISSISSIPPI  21.2608696  3.00000000   18.26086957
## 44          MISSOURI    23.0175439 -0.50000000   23.51754386
## 45          NEW MEXICO   33.0833333  2.00000000   31.08333333
## 46          ALABAMA     36.5303030  3.20000000   33.33030303
## 47          MONTANA     58.0000000  2.00000000   56.00000000
## 48 NORTH DAKOTA      59.0000000  0.00000000   59.00000000
## 49 SOUTH DAKOTA    297.0000000  7.00000000  290.00000000
## 50          WYOMING    614.0000000 16.00000000  598.00000000
## 51 WEST VIRGINIA 1268.0000000 19.00000000 1249.00000000
```

ACS, state

```
trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv") %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)
```

```
## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
```



```

pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
mutate(diff_euca = `0`-`-1`) %>%
mutate(region = "EUCA") %>%
select(diff_euca, state)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
mutate(diff_noneuca = `0`-`-1`) %>%
mutate(region = "Non-EUCA") %>%
select(diff_noneuca, state) %>%
merge(trajectory_agg_euca) %>%
mutate(diff = diff_noneuca - diff_euca) %>%
arrange(diff)

```

##	state	diff_noneuca	diff_euca	diff
## 1	MASSACHUSETTS	-0.08757896	-0.03928548	-0.04829349
## 2	CONNECTICUT	0.05456573	0.01496558	0.03960015
## 3	OKLAHOMA	0.05358913	0.00000000	0.05358913
## 4	HAWAII	0.25668549	0.19206146	0.06462403
## 5	OREGON	0.11358857	0.02351447	0.09007410
## 6	WASHINGTON	0.03639143	-0.05922241	0.09561384
## 7	NEW YORK	0.14797586	0.01706095	0.13091491
## 8	VERMONT	0.13803575	0.00000000	0.13803575
## 9	ALASKA	0.32599078	0.14099401	0.18499677
## 10	CALIFORNIA	0.31875618	0.08599806	0.23275811
## 11	RHODE ISLAND	0.41862412	0.11969239	0.29893173
## 12	ILLINOIS	0.37996227	0.04170294	0.33825933
## 13	MAINE	0.32831019	-0.04316298	0.37147317
## 14	NEW JERSEY	0.48125127	0.09881801	0.38243327
## 15	MARYLAND	0.63204246	0.15785320	0.47418927
## 16	DELAWARE	0.33322584	-0.17146776	0.50469361
## 17	MINNESOTA	0.41263498	-0.10430610	0.51694108
## 18	PENNSYLVANIA	0.69802106	0.02115686	0.67686421
## 19	COLORADO	0.99913770	0.10395659	0.89518111
## 20	WISCONSIN	1.14656985	0.14855254	0.99801730
## 21	NEW HAMPSHIRE	1.09802008	0.06964031	1.02837977
## 22	MICHIGAN	1.39282402	0.03090999	1.36191403
## 23	VIRGINIA	1.90949953	0.26337015	1.64612938
## 24	NEVADA	2.39171769	0.42620855	1.96550915
## 25	NORTH CAROLINA	2.38699405	0.30004251	2.08695155
## 26	NEBRASKA	2.34709869	0.25665155	2.09044713
## 27	NORTH DAKOTA	2.10782037	0.00000000	2.10782037
## 28	ARIZONA	2.17943654	0.03005598	2.14938056
## 29	GEORGIA	2.82259285	0.13880214	2.68379071
## 30	IDAHO	2.92589649	0.22962113	2.69627536
## 31	OHIO	2.87934897	0.13639774	2.74295124
## 32	DISTRICT OF COLUMBIA	3.46001451	0.16757904	3.29243547
## 33	TEXAS	4.01025626	0.65946107	3.35079519
## 34	FLORIDA	3.83652633	0.46049874	3.37602759
## 35	IOWA	3.69614769	0.16820858	3.52793912
## 36	KENTUCKY	4.21056341	0.10232273	4.10824068
## 37	MONTANA	4.44614795	0.16283993	4.28330802
## 38	KANSAS	4.73003913	0.42058777	4.30945136

## 39	INDIANA	4.93624369	0.52258617	4.41365752
## 40	UTAH	5.41501150	0.39247324	5.02253826
## 41	LOUISIANA	5.58903504	0.20524398	5.38379105
## 42	MISSOURI	6.09719260	-0.07309808	6.17029068
## 43	MISSISSIPPI	7.96092796	0.65645514	7.30447282
## 44	ARKANSAS	8.52153532	0.45042791	8.07110742
## 45	TENNESSEE	9.03235832	0.27200839	8.76034993
## 46	SOUTH DAKOTA	10.10754152	1.19986287	8.90767865
## 47	SOUTH CAROLINA	9.64415371	0.20479563	9.43935808
## 48	NEW MEXICO	11.21468927	0.21640338	10.99828589
## 49	ALABAMA	14.63571576	0.58597326	14.04974249
## 50	WYOMING	40.28607047	3.12317002	37.16290045
## 51	WEST VIRGINIA	55.53122537	2.47363625	53.05758912